

Sylow Theorems

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1 Group Actions

Let G be a finite group with order power of p act on a finite set X . Let $X_G = \{x \in X : gx = x \ \forall g \in G\}$. Then $|X| \equiv |X_G| \pmod{p}$.

Proof: X splits into G -orbits, so $|X| = |X_G| + \sum_{o \in \mathcal{O}, |o| > 1} |o|$. By Orbit-stabilizer, $|o||G_x| = |G|$ for some representative $x \in o$, so $|o| \mid |G|$. Since $|G|$ is a power of p and $|o| > 1$, $p \mid |o|$. Hence $|X| \equiv |X_G| \pmod{p}$.

Let G be a finite group with $p \mid |G|$. Then there exists an element $g \in G$ with order p .

Proof: I proceed by induction. If $|G| = p$, then $G \cong \mathbb{Z}_p$, for which any generator has order p .

Assume that the result holds for $|G| = jp$ for all $j \leq k$. Consider a group with $|G| = (k+1)p$. By the class equation,

$$|G| = |Z(G)| + \sum_{c \in \mathcal{C}^*} |c|.$$

Consider $c \in \mathcal{C}^*$ and pick $x \in c$. By Orbit Stabilizer, $|c||G_x| = |G|$. If $p \nmid |G_x|$, then by the IH, we're done, since $|G_x| < |G|$ and $|G_x| \mid |G|$.

Else, $p \mid |c|$ for all $c \in \mathcal{C}^*$, so $p \mid |Z(G)|$. By classification theorem for finitely generated Abelian groups, we're done.

Corr: Let G be finite. Then G is a p -group iff $|G|$ is a power of p .

Let G be a finite group, $H < G$ a p -group. Then $[N(H) : H] \equiv [G : H] \pmod{p}$.

Proof: Consider the action of H on the left cosets G/H . H is a finite p -group and therefore $|H|$ is a power of p . We have

$$h(gH) = gh \ \forall h \in H \iff g^{-1}hg \in H \ \forall h \in H \iff g^{-1}Hg = H.$$

Hence the fixed points of this action are those of $N(H)/H$, so $[N(H) : H] \equiv [G : H] \pmod{p}$.

2 Sylow Theorems

First Sylow theorem. Suppose G is a finite group with $|G| = p^k r$ with $p \nmid r$. Then there exists a chain of p -subgroups H_i such that

$$H_0 \triangleleft H_1 \triangleleft H_2 \cdots \triangleleft H_k,$$

with $|H_i| = p^i$, $H_{i+1}/H_i \cong \mathbb{Z}_p$ for $i \in [k-1]$.

Proof: $H_0 = e$ is the identity. Suppose the group H_i for $i \in [k-1]$ is defined. $[N(H_i) : H_i] \equiv [G : H_i] \equiv |G|/|H_i| \equiv 0 \pmod{p}$. $N(H_i)/H_i$ has group structure, so there exists a subgroup of order p . By the Lattice theorem, there exists a subgroup H_{i+1} such that $H_i < H_{i+1} < N(H_i)$ and $H_{i+1}/H_i \cong \mathbb{Z}_p$. Note that $H_i \triangleleft H_{i+1}$ as well.

By induction, $H_0, H_1, \dots, H_k$ are defined.

Second Sylow theorem. Suppose H is a p -subgroup of G and P is a p -Sylow subgroup of G . Then there exists $g \in G$ such that $gHg^{-1} < P$. In particular, all p -Sylow subgroups are conjugate.

Proof: H acts on $X = G/P$ by left multiplication. Since H is a (finite) p -group, $|X| \equiv |X_0| \pmod{p}$. Since $|X| \not\equiv 0 \pmod{p}$, $|X_0| \not\equiv 0 \pmod{p}$. Hence, there is a fixed point $gP \in G/P$, ie $hgP = gP$ for all $h \in H$, which implies $g^{-1}Hg < P$.

Third Sylow theorem. Suppose $n = p^k r$. The number of p -Sylow subgroups N divides r and $N \equiv 1 \pmod{p}$.

Proof: G acts on the set X of p -Subgroups by conjugation. Pick $H_0 \in X$, since $X \neq \emptyset$. By the Second Sylow theorem, G acts transitively. By Orbit-stabilizer, $N \cdot |N(H_0)| = |G|$, so $N = [G : N(H_0)]$. Observe that

$$[G : N(H_0)][N(H_0) : H_0] = [G : H_0].$$

It follows $N \mid [G : H_0] = r$. Since H_0 is a p -subgroup, $[G : H_0] \equiv [N(H_0) : H_0] \pmod{p}$. Since H_0 is a p -Sylow subgroup, $p \nmid [G : H_0]$, so $N = [G : N(H_0)] \equiv 1 \pmod{p}$.

Suppose $H < G$. Is it true $G/H \longleftrightarrow G/gHg^{-1}$ or when?

3 Solvable and Simple Groups

Let G be a group and suppose there exists a chain

$$G = G_0 \triangleright G_1 \triangleright G_2 \triangleright \cdots \triangleright G_n = 1$$

such that for $i \in [[n-1]]$ that G_i/G_{i+1} is Abelian. Show that equivalently the derived series terminates

$$G = G^{(0)} \triangleright G^{(1)} \triangleright G^{(2)} \triangleright \cdots \triangleright G^{(m)} = 1.$$

I claim that for $i \in [[n]]$ that $G^{(i)} < G_i$. I proceed by induction. The base case is $G < G$. Suppose the result for $i < n$, so $G^{(i)} < G_i$. Then $G^{(i+1)} < [G_i, G_i]$. Since G_i/G_{i+1} is Abelian, $[G_i, G_i] < G_{i+1}$, so $G^{(i+1)} < G_{i+1}$, as desired.

In particular, $G^{(n)} < G_n = 1$, so the derived series terminates.

Conversely, suppose the derived series terminates. For $i \in [[m-1]]$, $G^{(i)}/G^{(i+1)} = G^{(i)}/[G^{(i)}, G^{(i)}]$ which is Abelian, AWD.